FIITJEE **JEE (Main)-2018**

ALL INDIA TEST SERIES

FULL TEST – II

Time Allotted: 3 Hours **Maximum Marks: 360**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section-A (01 to 30)** contains 30 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

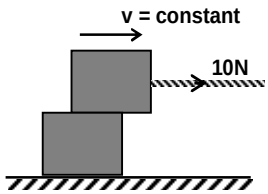
PART – I

SECTION – A
Straight Objective Type

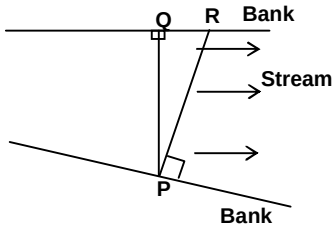
This section contains **30 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct.

1. A solid object is rotating freely without experiencing any external torque. In this case
   (A) both the angular momentum and angular velocity have constant direction.
   (B) the direction of the angular momentum is constant but the direction of the angular velocity might not be constant.
   (C) the direction of the angular velocity is constant but the direction of the angular momentum might not be constant.
   (D) neither the angular momentum nor the angular velocity necessarily has constant direction.

2. A 10-cm cube of metal is fastened rigidly in place. A second, identical cube of metal is pulled across the top of the first cube at a constant speed by a constant 10 N force, as shown. The frictional force between the cubes
   (A) is less than 10 N.
   (B) is equal to 10 N.
   (C) is greater than 10 N.
   (D) cannot be determined without a detailed model of the two surfaces.



3. In a lake, stream direction is as shown in the figure. A man starting from the point P on the bank 1 wants to move to the bank 2 in shortest time. He should swim
   (A) along PQ
   (B) along PR
   (C) In a direction in-between PQ and PR
   (D) in all cases he would reach at the same time



Space for rough work

4. A particle is projected vertically upwards from the ground with a speed v and a second particle is projected at the same instant from a height h directly above the first particle with the same speed v at an angle of projection θ with the horizontal in upward direction. The time when the distance between them is minimum is

(A) $\frac{h}{2v\sin\theta}$ (B) $\frac{h}{2v\cos\theta}$

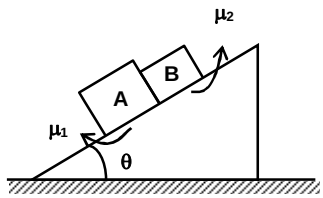
(C) $\frac{h}{\sqrt{2}\,v}$ (D) $\frac{h}{2v}$

5. Which of the following forces is non conservative one?

(A) $3\hat{i}+4\hat{j}$ (B) $4x\hat{i}+3y\hat{i}$

(C) $3x^2\hat{i}+4y^2\hat{j}$ (D) $y^2\hat{i}+x^2\hat{j}$

6. In the figure shown, the wedge is fixed and the masses are released from rest. The coefficient of friction between A and the incline is μ_1 and B and the incline is μ_2. Then which of the following sentences is (are) correct



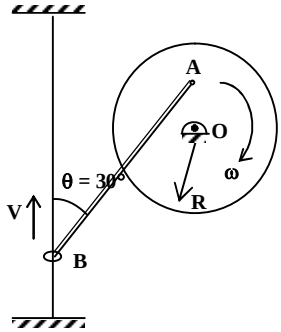
I. normal reaction between A and B can never be zero

II. normal reaction between A and B is zero only if $\mu_1 = \mu_2$

(A) I only (B) II only

(C) both I and II (D) none

7. A disk is fixed at its centre O and rotating with constant angular velocity ω. There is a rod whose one end is connected at A on the disc and the other end is connected with a ring which can freely move along the fixed vertical smooth rod. At an instant when the rod is making an angle 30° with the vertical the ring is found to have a velocity v in the upward direction. Find ω of the disk. Given that the point A is R/2 distance above point O and length of the rod AB is ℓ

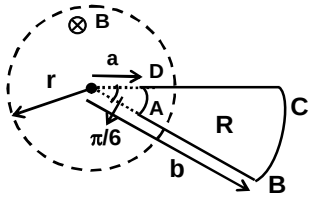


(A) $\frac{v\sqrt{3}}{2\ell}$ (B) $\frac{v\sqrt{3}}{2R}$

(C) $\frac{2\sqrt{3}v}{R}$ (D) $\frac{2v}{\ell\sqrt{3}}$

Space for rough work

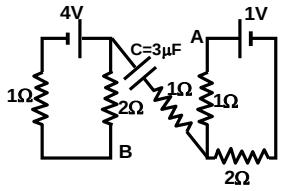
8. Magnetic field B in a cylindrical region of radius r varies according to the law $B = B_0t$ as shown in the figure. A fixed conducting loop ABCDA of resistance R is lying in the region as shown. The current flowing through the loop is



(A) $\frac{\pi a^2 B_0}{12R}$ (B) $\frac{\pi\left(r^2 - a^2\right)B_0}{12R}$

(C) $\frac{\pi\left(b^2 - a^2\right)B_0}{12R}$ (D) none of the above

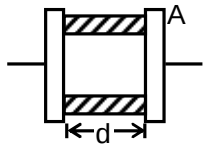
9. In the circuit shown in the adjacent figure, the batteries are ideal. The charge on the capacitor C is



(A) 10 μC (B) 20 μC

(C) 30 μC (D) zero

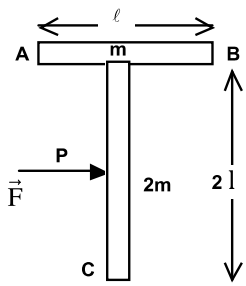
10. A parallel plate capacitor of area A and separation d is provided with thin insulating spacers to keep its plates aligned in an environment of fluctuating temperature. If the coefficient of thermal expansion of material of the plate is α, find the coefficient of thermal expansion (α_s) of the spacers in order that the capacitance does not vary with temperature. (Ignore effect of the spacers on capacitance.)



(A) $\alpha_s = \frac{\alpha}{2}$ (B) $\alpha_s = 3\alpha$

(C) $\alpha_s = 2\alpha$ (D) $\alpha_s = \alpha$

11. A T shaped object with dimensions shown in the figure is lying on a smooth floor. A force $\vec{F}$ is applied at the point P parallel to AB such that the object has only the translational motion without rotation. Find the location of P with respect to C.

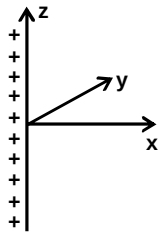


(A) $\frac{3}{4}\ell$ (B) ℓ

(C) $\frac{4}{3}\ell$ (D) $\frac{3}{2}\ell$

Space for rough work

12. An infinite thread of charge density λ lies along z-axis. The potential difference between points A (4, 3, 4) and B (3, 4, 0) is



(A) $\frac{\lambda}{2\pi \in_0} \ln\left(\frac{\sqrt{41}}{5}\right)$ (B) $\frac{\lambda}{2\pi \in_0} \ln(5)$

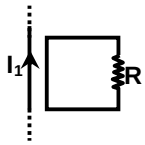
(C) zero (D) $\frac{\lambda}{2\pi \in_0}$

13. The sun having surface temperature T_S radiates like a black body. The radius of sun is R_S and earth is at a distance R from the surface of sun. Earth absorbs radiations falling on its surface from sun only and is at constant temperature T. If radiations falling on earth's surface are almost parallel and earth also radiates like a blackbody, then

(A) $T = T_S\sqrt{\frac{R_S}{2R}}$ (B) $T = T_S$

(C) $T = \frac{T_S}{2}\sqrt{\frac{R_S}{R}}$ (D) $T = T_S\sqrt{\frac{R_S}{R}}$

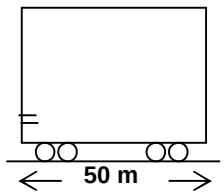
14. In the adjacent figure, the mutual inductance of the infinite straight wire and the coil is M, while the self inductance of the coil is L. The current in infinite wire is varying according to the relation $I_1 = \alpha t$, where α is a constant and t is the time. The time dependence of current in the coil is



(A) $\frac{M\alpha}{R}$ (B) $\frac{M\alpha L}{R}e^{-Rt/L}$

(C) $\frac{\alpha}{R}\left(1-e^{-Rt/L}\right)$ (D) none of the above

15. A cannon-ball is fired with a speed 50 m/sec with respect to ground from one end of a stationary railroad car which rests on a smooth track. The ball collides with the opposite wall and stops. How much distance did the railroad car cover just before the collision? Given that the ratio of the mass of the ball to that of the railroad car is 1/99 and the length of the car is 50 m



(A) 2 m

(B) $\frac{1}{2}$m

(C) 0 m

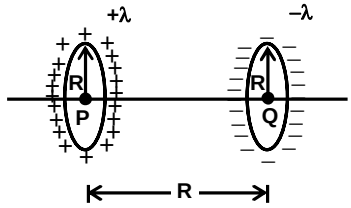
(D) cannot be calculated from acts

Space for rough work

16. A cubical box of side 1 m contains an ideal gas at pressure 100 N/m^2. If $\Sigma v_x^2 = \Sigma v_y^2 = \Sigma v_z^2 = 10^{28}\ m^2/s^2$, where v_x, v_y and v_z are the x, y and z components, respectively, of the gas molecule, then the mass of each gas molecule is

(A) 10^{-20} g (B) 10^{-23} g
(C) 10^{-18}g (D) 10^{-26} g

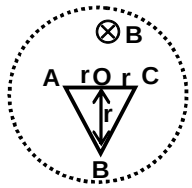
17. Two fixed insulating rings A and B carry charges with uniform linear charge density $+\lambda$ and $-\lambda$, respectively, as shown in the adjacent figure. The planes of the rings are parallel to each other and their axes are coinciding. A particle of charge "q" and mass "m" is released with zero velocity from centre P of the positively charged ring. The kinetic energy of the particle when it reaches centre Q of the negatively charged ring will be



(A) $\sqrt{\frac{2\lambda q}{m\in_0}\left(1-\frac{1}{\sqrt{2}}\right)}$ (B) $\frac{\lambda}{\in_0}\left(1-\frac{1}{\sqrt{2}}\right)q$

(C) $\frac{\lambda}{2\in_0}\left(1-\frac{1}{\sqrt{2}}\right)q$ (D) none of the above

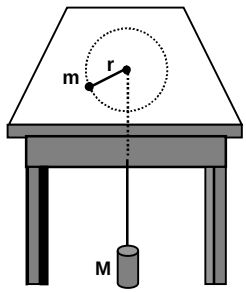
18. Magnetic field in the cylindrical region with its axis passing through O varies at a constant rate $\frac{dB}{dt}$. A triangular imaginary loop ABC, with AB = BC, is lying in this region as shown in the adjacent figure. The work done to move unit positive charge from A to B along the side AB is



(A) $r^2\frac{dB}{dt}$ (B) $\pi r^2\frac{dB}{dt}$

(C) $\frac{r^2}{2}\frac{dB}{dt}$ (D) $\frac{3r^2}{4}\frac{dB}{dt}$

19. A puck is moving in a circle of radius r_0 with a constant speed v_0 on a level frictionless table. A string is attached to the puck, which holds it in the circle; the string passes through a frictionless hole and is attached on the other end to a hanging object of mass M. The puck is now made to move with a speed $v' = 3v_0$, but still in circle. The mass of the hanging object is left unchanged. The acceleration a' of the puck and the radius r' of the circle are now given by

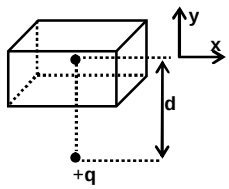


(A) $a' = 9a_0$ and $r' = r_0$. (B) $a' = a_0$ and $r' = 9r_0$
(C) $a' = 9a_0$ and $r' = 9r_0$ (D) $a' = a_0$ and $r' = r_0$

Space for rough work

20. Two spherical black bodies A and B are emitting radiations at same rate. The radius of B is doubled keeping radius of A fixed. The wavelength corresponding to maximum intensity becomes half for A while it remains same for B. Then, the ratio of rate of radiation energy emitted by A and B is
(A) 2 (B) 1/2
(C) 4 (D) 1/4

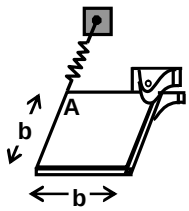
21. A charge +q is placed at a distance 'd' from the centre of the uncharged metallic cube of side 'a'. The electric field at the centre of the cube due to induced charges on the cube will be



(A) zero (B) $\frac{q}{4\pi \in_0 d^2}\left(-\hat{j}\right)$
(C) $\frac{q}{4\pi \in_0 d^2}\left(\hat{j}\right)$ (D) $\frac{q}{4\pi \in_0 \left(d-\frac{a}{2}\right)^2}\left(-\hat{j}\right)$

22. A physical quantity ρ is calculated by using the formula $\rho = \frac{1}{10}\frac{xy^2}{z^{1/3}}$, where x, y and z are experimentally measured quantities. If the fractional error in the measurement of x, y and z are 2%, 1% and 3%, respectively, then the maximum fractional error in the calculation of ρ is
(A) 0.5% (B) 5%
(C) 6% (D) 7%

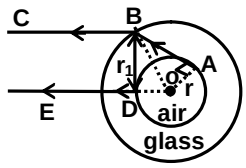
23. A uniform square plate of mass m is supported in a horizontal plane by a vertical pin at B and is attached at A to a spring of constant K. If corner A is given a small displacement and released, determine the period of the resulting motion.



(A) $2\pi\sqrt{\frac{2Mb}{3K}}$ (B) $2\pi\sqrt{\frac{Mb}{6K}}$
(C) $\pi\sqrt{\frac{Mb}{3K}}$ (D) none of the above

Space for rough work

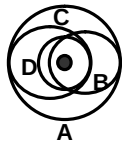
24. The adjacent figure shows cross section of a hollow glass tube of internal radius r, external radius R and index of refraction n. For two rays DE and ABC (in which DE lies on ODE and DE and BC are parallel), the separation r_1 will be



(A) $r_1 = (n - 1)R$ (B) $r_1 = n^2R$

(C) $r_1 = nr$ (D) $r_1 = n^2r$

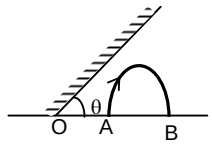
25. The given figure shows several possible elliptical orbits of a satellite. On which orbit will the satellite acquire the largest speed?



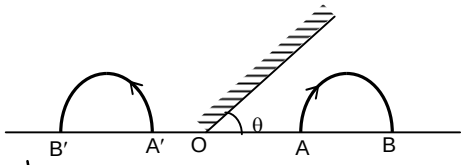
(A) A (B) B

(C) C (D) D

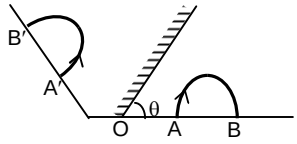
26. An object is projected from A and reaches B on the same horizontal surface. A plane mirror inclined at an angle θ to the horizontal is placed on the ground as shown in the figure. Which of the following best describes the path of the image?



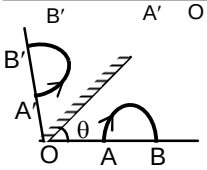
(A)



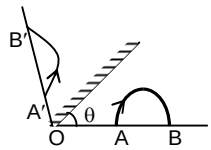
(B)



(C)



(D)



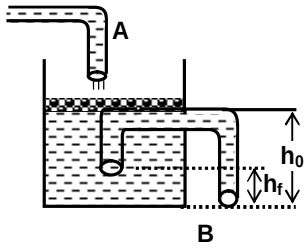
27. A solid uniform ball of volume V floats on the interface of two immiscible liquids [The specific gravity of the upper liquid is $\gamma/2$ and that of the lower liquid is 2γ, where γ is the specific gravity of the solid ball.] The fraction of the volume of the ball that will be in upper liquid is

(A) $\frac{2}{3}V$ (B) $\frac{1}{3}V$

(C) $\frac{V}{4}$ (D) $\frac{3V}{4}$

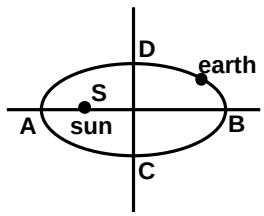
Space for rough work

28. In the figure shown, water flows into the open vessel through pipe A at a constant rate. The water can flow out of the vessel by pipe B. The maximum flow-rate of the water in pipe B is slightly more than that in pipe A. Final height of the water level in the vessel would be



(A) h_0
(B) h_f
(C) somewhere in-between h_0 and h_f
(D) The water level will keep oscillating in the vessel

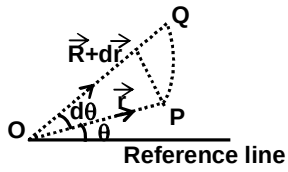
29. The earth is moving on an elliptical path, whose one focus sun is situated as shown in figure. If AS = r_{min} and SB = r_{max}, the sun and earth system obey the Kepler's law, the square of time-period is directly proportional to



(A) $(r_{min})^3$ (B) $(r_{max})^3$

(C) $\left(\frac{r_{min}+r_{max}}{2}\right)^3$ (D) $\left(\frac{r_{min}r_{max}}{r_{min}+r_{max}}\right)^3$

30. A body is moving under the action of central force $\vec{F}(r)\hat{e}_r$ such that its position vector is $\vec{r}=r\hat{e}_r$. Then, choose the correct statement (symbols are having usual meaning and $\hat{e}_r$, $\hat{e}_\theta$ denote unit vectors along the radial and tangential direction, respectively) from the following.



(A) $\vec{v}=\frac{dr}{dt}\hat{e}_r+r\frac{d\theta}{dt}\hat{e}_\theta,\ \vec{a}=\left[\frac{d^2r}{dt^2}-r\left(\frac{d\theta}{dt}\right)^2\right]\hat{e}_r,\ 2\frac{dr}{dt}\frac{d\theta}{dt}+r\frac{d^2\theta}{dt^2}=0$

(B) $\vec{v}=\frac{dr}{dt}\hat{e}_r+r\frac{d\theta}{dt}\hat{e}_\theta,\ \vec{a}=\left[\frac{d^2r}{dt^2}+r\left(\frac{d\theta}{dt}\right)^2\right]\hat{e}_r,\ 2\frac{dr}{dt}\frac{d\theta}{dt}-r\frac{d^2\theta}{dt^2}=0$

(C) $\vec{v}=\frac{dr}{dt}\hat{e}_r-r\frac{d\theta}{dt}\hat{e}_\theta,\ \vec{a}=\left[\frac{d^2r}{dt^2}+r\left(\frac{d\theta}{dt}\right)^2\right]\hat{e}_r,\ 2\frac{dr}{dt}\frac{d\theta}{dt}+r\frac{d^2\theta}{dt^2}=0$

(D) $\vec{v}=\frac{dr}{dt}\hat{e}_r-r\frac{d\theta}{dt}\hat{e}_\theta,\ \vec{a}=\left[\frac{d^2r}{dt^2}-r\left(\frac{d\theta}{dt}\right)^2\right]\hat{e}_r,\ 2\frac{dr}{dt}\frac{d\theta}{dt}+r\frac{d^2\theta}{dt^2}=0$

Space for rough work

Chemistry

PART – II

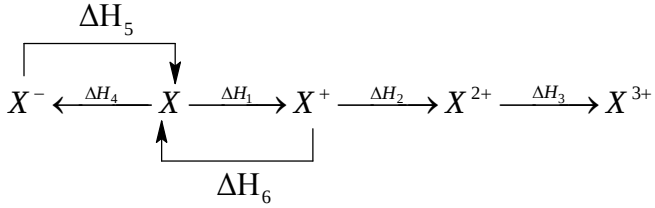
SECTION – A
Straight Objective Type

This section contains **30 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct.

31. The following electronic transition occurs when lithium atoms are sprayed into a hot flame;
$2s \xrightarrow{I} 2p \xrightarrow{II} 3d \xrightarrow{III} 3p \xrightarrow{IV} 4s \xrightarrow{V} 3p,$
Which of the transition would result in the emission of light?
(A) I, II and IV (B) III and V
(C) III, IV and V (D) All of these steps

32. The rate law for a reaction between the substances A and B is given by rate $= k[A]^n[B]^m$. On doubling the concentration of A and halving the concentration of B, the ratio of the new rate to the earlier rate of the reaction will be:
(A) $1/2^{m+n}$ (B) $2^{(m+n)}$
(C) $1/2^{n-m}$ (D) $2^{(n-m)}$

33. Consider the following sequence of reaction:



If electronic configuration of element X is $[Ne]3s^1$, then which of the following order is incorrect regarding given enthalpies?
(A) $|\Delta H_4| = |\Delta H_5|$ (B) $|\Delta H_2| > |\Delta H_1|$
(C) $|\Delta H_2| > |\Delta H_3|$ (D) $|\Delta H_1| = |\Delta H_6|$

Space for rough work

34. The bond angle of Cl_2O is:

(A) Smaller than that of F_2O (B) Greater than that of H_2O

(C) Smaller than that of H_2O (D) Same as that of F_2O

35. For the reaction: $2NO(g)+Cl_2(g) \rightleftharpoons 2NOCl(g)$, NO and Cl_2 are initially taken in mole ratio $2:1$. The total pressure at equilibrium is found to be 1 atm. If mole of $NOCl$ are one fourth of that of Cl_2 at equilibrium, K_p for the reaction is:

(A) 13/36 (B) 13/256

(C) 13/512 (D) 13/128

36. Suppose that in any Bohr atom or ion orbits are only in even numbers like $2,4,6.....$ The maximum wavelength of radiation emitted in the visible region of hydrogen spectrum should be:

(A) $4/R$ (B) $R/4$

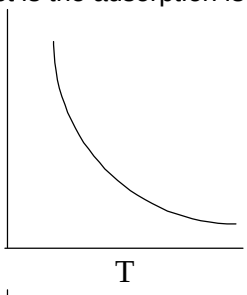
(C) $36/5R$ (D) $16/3R$

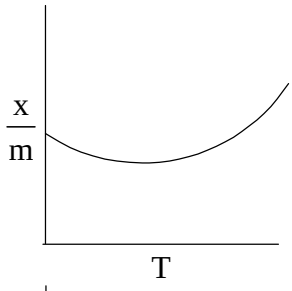
37. The overall charge present on the cyclic silicate anion $[Si_6O_{18}]^{n-}$ is:

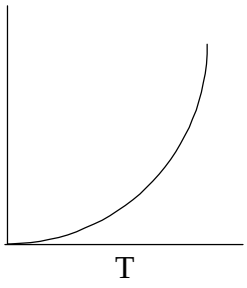
(A) 6 (B) 12

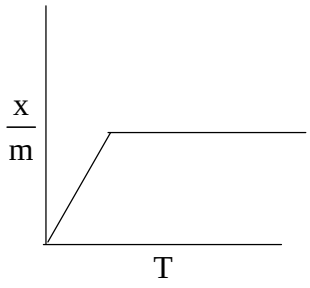
(C) 18 (D) 24

38. Which plot is the adsorption isobar for chemisorption?

(A) 

(B) 

(C) 

(D) 

Space for rough work

39. The solubility product of PbI_2 is 7.2×10^{-9}. The maximum mass of NaI which may be added in $500ml$ of $0.005M$ $Pb(NO_3)_2$ solution without any precipitation of PbI_2 is $(I=127)$:

(A) $0.09g$ (B) $1.2\times10^{-3}g$

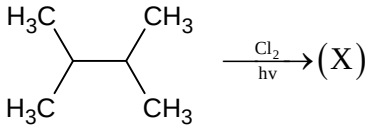
(C) $6\times10^{-4}g$ (D) $1.08\times10^{-5}g$

40. The number of moles of $Cr_2O_7^{2-}$ needed to oxidise 0.136 equivalent of $N_2H_5^+$ through the reaction:

$N_2H_5^+ + Cr_2O_7^{2-} \longrightarrow N_2 + Cr^{3+} + H_2O$ is:

(A) 0.0226 (B) 0.087

(C) 0.136 (D) 0.488

41. 

Mono chloro products: (including stereoisomers)

(A) 2 (B) 3

(C) 4 (D) 5

42. Which of the following reaction is non-redox?

(A) $2NaNO_3 \longrightarrow 2NaNO_2 + O_2$

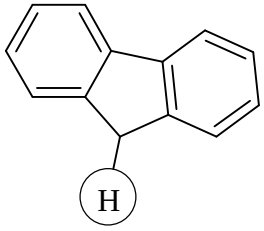
(B) $KCl + NaNO_3 \longrightarrow KNO_3 + NaCl$

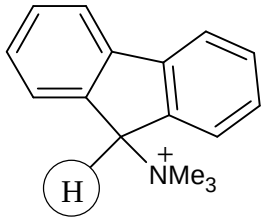
(C) $Fe + H_2SO_4 \longrightarrow FeSO_4 + H_2$

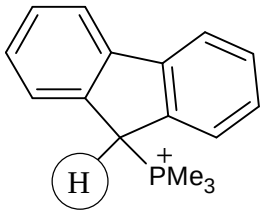
(D) $4Ag + 8CN^- + O_2 + H_2O \longrightarrow 4\left[Ag(CN)_2\right]^- + 4OH^-$

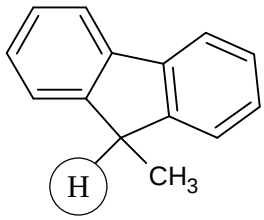
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43. Which of the following circled hydrogen is most acidic?

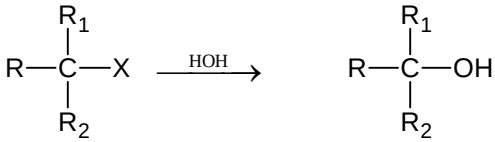
(A) 

(B) 

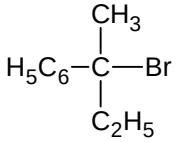
(C) 

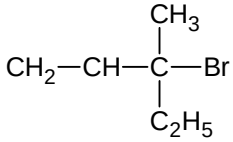
(D) 

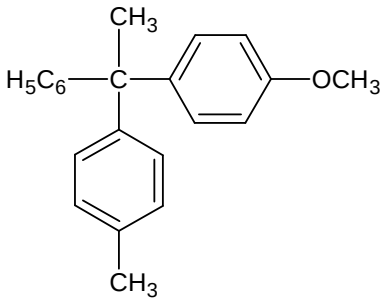
44. For the given reaction,

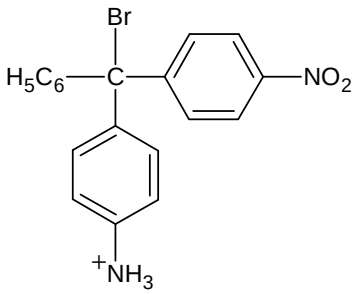


Which substrate will give maximum racemisation?

(A) 

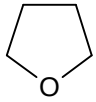
(B) 

(C) 

(D) 

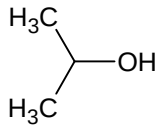
Space for rough work

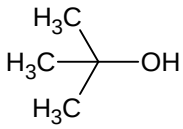
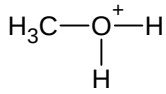
45. Compare the rate of reaction of following species with OH^- :

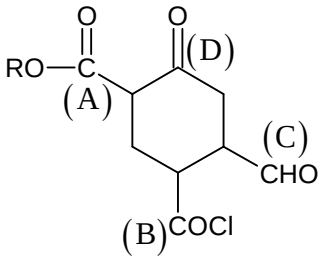
(i)  (ii)  (iii) 

(A) i>ii>iii (B) i>iii>ii
(C) iii>ii>i (D) ii>i>iii

46. Which of the following is stronger acid than methanol?

(A) $CH_3 - C \equiv CH$ (B) 

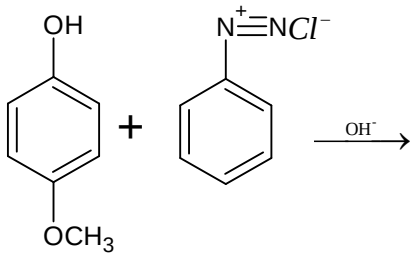
(C)  (D) 

47. 

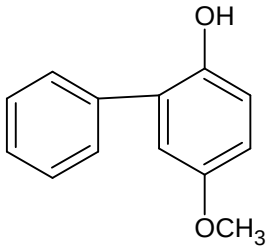
Find the correct order of reactivity of carbonyl groups with $LiAlH_4$:

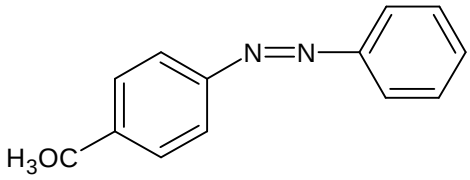
(A) A>B>C>D (B) B>C>D>A
(C) D>C>B>A (D) B>D>C>A

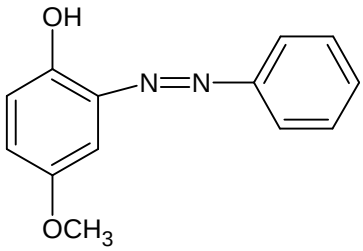
Space for rough work

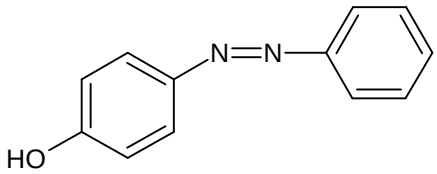
48. 

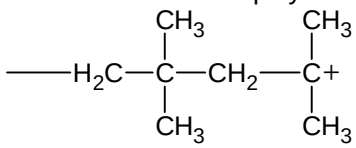
Product of reaction is:

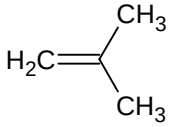
(A) 

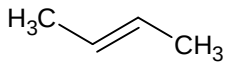
(B) 

(C) 

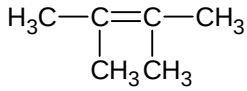
(D) 

49. The monomer of the polymer  is:

(A) 

(B) 

(C)

(D) 

Space for rough work

50. The correct order of matching of complex compound in column I with the properties in column II:

	Column I		Colum II
(A)	$\left[Cr\left(NH_3\right)_6\right]^{3+}$	(P)	Tetrahedral and paramagnetic
(B)	$\left[Co\left(CN\right)_6\right]^{3-}$	(Q)	Octahedral and diamagnetic
(C)	$\left[Ni\left(CN\right)_4\right]^{2-}$	(R)	Octahedral and paramagnetic
(D)	$\left[Ni\left(Cl\right)_4\right]^{2-}$	(S)	Square planar and diamagnetic

(A) A-R, B-Q, C-P, D-S (B) A-Q, B-R, C-P, D-S

(C) A-R, B-Q, C-S, D-P (D) A-Q, B-R, C-S, D-P

51. Which of the following statements ore incorrect regarding roasting?
(A) Impurities are removed in the form of their elemental vapours
(B) Lower oxidation states are oxidised further
(C) Sulphide ores are converted to their oxides
(D) The temperature of the process is maintained just above the melting point of mixture

52. A sulphide of a metal $\left(M\right)$ with formula MS is white coloured ppt which is soluble in acidic solution $\left(PH<6\right)$. Metal cation M is:

(A) Pb^{2+} (B) Zn^{2+}

(C) Hg^{2+} (D) Cu^{2+}

53. Which of the following is not correct regarding the diagonal relationship between Al and Be?
(A) Oxides of both are amphoteric in nature
(B) Carbides of both produce same gas an hydrolysis
(C) Hydrides of both are electron deficient covalent compounds
(D) Both show +2 oxidation states in their compounds

54. The final product obtained from the electrolysis of 50% H_2SO_4 with high current density has:

(A) S-O-O-S linkage (B) S=S linkage

(C) S-S linkage (D) S-O-S linkage

55. The gradual decrease in atomic radius for lanthanoids elements in not obeyed by:

(A) Eu only (B) Yb only

(C) Both Eu & Yb (D) None of these

Space for rough work

56. A fixed mass of an ideal gas contained in a 24.63L sealed rigid vessel at 1 atm is heated from $-73°C$ to $27°C$. Calculate change in Gibb's energy if entropy of gas is a function of temperature as $S=2+10^{-2}T\left(J/k\right)$: [Use 1atm L = 0.1 kJ]

(A) $1231.5J$ (B) $1281.5J$

(C) $781.5J$ (D) 0

57. The dissociation constant of a weak acid is 1.6×10^{-5} and molar conductivity at infinite dilution is $380\times10^{-4}\ Sm^2mol^{-1}$. If the cell constant is $0.01m^{-1}$ the conductance of $0.01M$ acid solution is:

(A) $1.52\times10^{-5}\ S$ (B) $1.52\ S$

(C) $1.52\times10^{-3}\ S$ (D) $1.52\times10^{-4}\ S$

58. CdO has $NaCl$ like structure with density $8.27g/cc$. If the ionic radius of O^{2-} is $1.24\,A°$, determine the ionic radius of Cd^{2+}:

(A) $1.5A°$ (B) $1.1A°$

(C) $1.9A°$ (D) $1.5A°$

59. Which of the following is a character of a gas at Boyle temperature?

(A) The effects of the repulsive and attractive intermolecular forces offset each other

(B) The repulsive intermolecular forces are stronger than the attractive intermolecular forces

(C) The repulsive intermolecular forces are weaker than attractive forces

(D) $b-\frac{a}{RT}>0$

60. An unknown oxide of manganese is reacted with carbon to form manganese metal and CO_2. Exactly $31.6g$ of the oxide, Mn_xO_y, yielded $13.2g$ of CO_2. The simplest formula of the oxide is $\left(Mn=55\right)$:

(A) MnO (B) MnO_2

(C) Mn_2O_3 (D) Mn_3O_4

Space for rough work

Mathematics

PART – III

SECTION – A
Straight Objective Type

This section contains **30 multiple choice questions**. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct.

61. $\sum_{\lambda=1}^{10} \sin^{-1}\left(\sin\left(\lambda\pi - \frac{\pi}{6}\right)\right)$ is equal to-

(A) $\frac{5\pi}{3}$ (B) $\frac{\pi}{2}$

(C) 0 (D) 5π

62. If $f(x) = \frac{\sin\frac{\pi x}{4}}{x+1}$, then $\lim_{h\to 0} \frac{f(1+h)-f(1)}{h^2+2h}$ is-

(A) $\frac{\pi-4}{2\sqrt{2}}$ (B) $\frac{\pi}{16\sqrt{2}}$

(C) $\frac{\pi-2}{8\sqrt{2}}$ (D) $\frac{\pi-2}{16\sqrt{2}}$

63. Number of solution of the equation $2^x + x = 2^{\sin x} + \sin x$ in $[0,10\pi]$ is-

(A) 5 (B) 6

(C) 11 (D) 1

64. Let ABCD is a parallelogram where $\overrightarrow{AB}=\vec{a}, \overrightarrow{AD}=\vec{b}, |\vec{a}|=|\vec{b}|=2$ and $|\vec{a}\times\vec{b}|+\vec{a}.\vec{b}=\sqrt{2}|\vec{a}||\vec{b}|\left(\vec{a}.\vec{b}>0\right)$, then area of this parallelogram, is (in square units)-

(A) $2\sqrt{2}$ (B) 2

(C) $\sqrt{2}$ (D) $8\sqrt{2}$

Space for rough work

65. Let the volume of tetrahedron ABCD is 81 cubic & G_1,G_2,G_3 are centroid of triangular faces ABC, ABD & ACD respectively, then volume of tetrahedron $AG_1G_2G_3$,is(in cubic units)

(A) 3 (B) 6

(C) $\frac{81}{4}$ (D) 54

66. Equation of plane containing the line $x+2y+3z-5=0=3x+2y+z-5$ which is parallel to line which is parallel to line $x-1=2-y=z-3$, is-

(A) $x+2y+z=0$ (B) $x-z=0$

(C) $2y+2z=1$ (D) $x+4y+3z=0$

67. Let A & B are two non singular matrices of order 3 such that $A+B=I\&A^{-1}+B^{-1}=2I$,then $\left|adj(4AB)\right|$, is (where adj (A) is adjoin of matrix A)-

(A) 4 (B) 16

(C) 64 (D) 128

68. If $r,k,p \in W$, then $\sum_{r+k+p=10} {}^{30}C_r,{}^{20}C_k.{}^{10}C_p$ is equal to-

(A) $\binom{60}{50}$ (B) $\binom{60}{30}$

(C) $\binom{60}{20}$ (D) $\binom{30}{10}\binom{30}{20}$

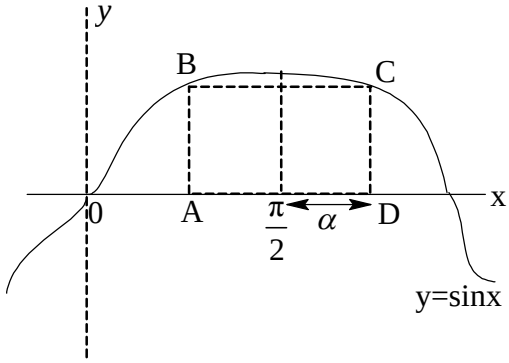
69. Function $f(x)=\begin{cases}\text{sgn}\left(\left[x\right]\right);x\neq I\\ \left[\text{sgn}\left(x\right)\right];x=I\end{cases}$ is {where sign () donates signum function & [.] denotes greatest integer function}

(A) Continuous but not differentiable at x=2 (B) Discontinuous at every integer

(C) Non differentiable at x=0 & 1 only (D) Non differentiable at every natural number

Space for rough work

70. A rectangle ABCD is inscribed in the region bounded by $y = \sin x$, x-axis where $x \in [0, \pi]$ (as shown in figure), then area of rectangle is maximum when 'α' satisfies



(A) $\tan\alpha=\alpha$ (B) $\cot\alpha=\alpha$
(C) $\sin\alpha=\alpha$ (D) $\cos\alpha=\alpha$

71. If $f(0) = 2$, then $\lim\limits_{x\to 0} \dfrac{\int_0^x \left(tf(x) + xf(t)\right) dt}{x^2}$ is equal to-

(A) 1 (B) 2
(C) 3 (D) 4

72. If $f(x)$ & $g(x)$ are inverse function of each other such that $f(1)=3$ & $f(3)=1$, then $\int_1^3 \left(g(x) + \dfrac{x}{f'(g(x))} \right) dx$ is equal to-

(A) 0 (B) 8
(C) -8 (D) 3

73. If solution of differential equation $\dfrac{dy}{dx} = \dfrac{1+x}{2y}$ is a conic passing through point $(1,1)$, then its eccentricity, is-

(A) 0 (B) $\sqrt{\dfrac{3}{2}}$
(C) 1 (D) $\sqrt{\dfrac{5}{3}}$

Space for rough work

74. Let tangents drawn from point $C(0,-b)$ to hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ touches hyperbola at points A & B. If ΔABC is a right angled triangle, then $\frac{a^2}{b^2}$ is equal to-

(A) 1 (B) $\frac{1}{2}$

(C) 2 (D) $\frac{3}{2}$

75. If normal at point P(t) to parabola $y^2=16x$ meets it again at point $Q(36,-24)$, then maximum possible focal distance of point P is-

(A) 8 (B) 16

(C) 32 (D) 20

76. If curves $y=ax^2+bx+c$ *and* $y=px^2+qx+r$ do not intersect each other and a a, b, c, p, q, r $\in\{1,2,3,4\ldots\ldots,10\}$ then maximum value of $(aq-bp)^2(c-r)^2$ is-

(A) 81 (B) 200

(C) 162 (D) 100

77. Let $n(A)=3, n(B)=3$ (where n(S) denotes number of elements in set S), then number of subsets of $(A\times B)$ having odd number of elements is-

(A) 64 (B) 128

(C) 256 (D) 512

78. Let A & B are two independent events such that $P(A)+P(B)=\frac{3}{4}$ & $P(\vec{A}/B)=\frac{2}{5}$, then $P(A\cap B)$ is-

(A) $\frac{9}{20}$ (B) $\frac{9}{100}$

(C) $\frac{5}{17}$ (D) $\frac{6}{20}$

Space for rough work

79. If complex numbers z_1 and z_2 both satisfy $z+\bar{z}=2|z-1|$ and $\arg(z_1-z_2)=\frac{\pi}{3}$, then value of $\text{Im}(z_1+z_2)$ is, where $\text{Im}(z)$ denotes imaginary part of z-

(A) $\sin\frac{\pi}{3}$ (B) $\text{cosec}\frac{\pi}{3}$

(C) $\tan\frac{\pi}{3}$ (D) $\cot\frac{\pi}{3}$

80. If ω is complex cube root of unit whose imaginary part is positive and $|z-\omega|=|z+\omega|$ arg(z) can be-

(A) $\frac{\pi}{3}$ (B) $\frac{5\pi}{6}$

(C) $\frac{\pi}{6}$ (D) $\frac{\pi}{4}$

81. A variable circle is drawn passing through the origin 'O'. It intersects X & Y axis respectively at point A & B such that OA+2OB=K (non zero constant), then circle always passes through a fixed point P other origin. P lies on-

(A) the x-axis (B) the line y=2x

(C) the line x=2y (D) the y-axis

82. Area bounded by curves $x=\sqrt{2-y^2}$ *and* $|x|=|y|$,is-

(A) $\frac{\pi}{2}$ (B) π

(C) $\frac{3\pi}{2}$ (D) 2π

83. The angle of elevation of a tower from point P due north of it is 30° and from a point Q due east of point P is θ. If distance of foot of tower from point P is 3m & $PQ=\sqrt{3}m$, then 'θ' is equal to-

(A) $\tan^{-1}2$ (B) $\cot^{-1}2$

(C) $\frac{\pi}{6}$ (D) $\frac{\pi}{4}$

Space for rough work

84. Let M denotes set of all 3×3 non singular matrices. Define the relation R by $R=\{(A,B)\in M\times M : AB=BA\}$, then R is-
(A) Reflexive, symmetric but not transitive
(B) Reflexive, symmetric & transitive
(C) Reflexive transitive but not symmetric
(D) Neither reflexive nor symmetric nor transitive

85. Interval in which function $y=\left|x^2-\left|x\right|-2\right|$ is non monotonic ,can be-
(A) $x\in(-2,-1)$
(B) $x\in(-4,-2)$
(C) $x\in(0,2)$
(D) $x\in(2,10)$

86. If $f(x)=x^4+\lambda x^3+x^2(\lambda\in R)$ has local maximum at $x=\frac{1}{2}$,then absolute minimum value of $f(x)$ is-
(A) -4
(B) 0
(C) 4
(D) -16

87. Let $x_1,x_2,.....,x_{100}$ are 100 observation such that $\sum x_i=0,\ \sum_{1\le i<j\le100}\left|x_ix_j\right|=80000$ &mean deviation from their mean is 5, then their standard deviation, is-
(A) 10
(B) 30
(C) 40
(D) 50

88. If statement $(p\to q)\to(q\to r)$ is false, then truth values of statement p,q,r respectively, can be-
(A) FTF
(B) TTT
(C) FFF
(D) FTT

89. Let $f(x)=\max(\sin x,\cos x)$
$g(x)=\min(\cos x,\sin x)$
$h(y)=f(x)y^2+ay+g(x)$
If equation $h(y)=0$ has roots $\forall x\in R$, then complete set of values of a is-
(A) $a\in\left(-\infty,-\sqrt{2}\right]\cup\left[\sqrt{2},\infty\right)$
(B) $a\in\left[-\sqrt{2},\sqrt{2}\right]$
(C) $a\in R$
(D) None of these

90. Let x_n,y_n,z_n,w_n denotes n^{th} terms of four different arithmetic progressions with positive terms. If $x_4+y_4+z_4+w_4=8$ and $x_{10}+y_{10}+z_{10}+w_{10}=20$, then maximum value of $x_{20}.y_{20}.z_{20}.w_{20}$ is-
(A) 10^4
(B) 10^6
(C) 10^8
(D) 10^{10}

Space for rough work